Exercise 7

Joshua John E. Santiago

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#RLab: Data Structures - Data Frames

#Part A — Creating Data Frames

#Question #1

```
# Part a: check that all six vectors have the same length
Respondent <- c(1:20)
Sex <- c(2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 1, 2)
FathersOccupation <- c(1, 3, 3, 3, 1, 2, 3, 1, 1, 1, 3, 2, 1, 3, 3, 1, 3, 1, 2, 1)
PersonsAtHome <- c(5, 7, 3, 8, 5, 9, 6, 7, 8, 4, 7, 5, 4, 7, 8, 8, 3, 11, 7, 6)
SiblingsAtSchool <- c(6, 4, 4, 1, 2, 1, 5, 3, 1, 2, 3, 2, 5, 5, 2, 1, 2, 5, 3, 2)
TypesOfHouses <- c(1, 2, 3, 1, 1, 3, 3, 1, 2, 3, 2, 3, 2, 2, 3, 3, 3, 3, 3, 2)

c(length(Respondent), length(Sex), length(FathersOccupation),
  length(PersonsAtHome), length(SiblingsAtSchool), length(TypesOfHouses))
```

```
## [1] 20 20 20 20 20 20
```

```
# Combine all six into a data frame
survey <- data.frame(Respondent = Respondent,
                     Sex = Sex,
                     FathersOccupation = FathersOccupation,
                     PersonsAtHome = PersonsAtHome,
                     SiblingsAtSchool = SiblingsAtSchool,
                     TypesOfHouses = TypesOfHouses)
print(survey)
```

```
##    Respondent Sex FathersOccupation PersonsAtHome SiblingsAtSchool
## 1           1   2                 1             5                6
## 2           2   2                 3             7                4
## 3           3   1                 3             3                4
## 4           4   2                 3             8                1
## 5           5   2                 1             5                2
## 6           6   2                 2             9                1
## 7           7   2                 3             6                5
## 8           8   2                 1             7                3
## 9           9   2                 1             8                1
## 10         10   2                 1             4                2
## 11         11   1                 3             7                3
## 12         12   2                 2             5                2
## 13         13   2                 1             4                5
## 14         14   2                 3             7                5
## 15         15   2                 3             8                2
## 16         16   2                 1             8                1
```

```
## 17         17   2                3              3                2
## 18         18   2                1             11                5
## 19         19   1                2              7                3
## 20         20   2                1              6                2
##    TypesOfHouses
## 1              1
## 2              2
## 3              3
## 4              1
## 5              1
## 6              3
## 7              3
## 8              1
## 9              2
## 10             3
## 11             2
## 12             3
## 13             2
## 14             2
## 15             3
## 16             3
## 17             3
## 18             3
## 19             3
## 20             2
```

```
# Part a: structure and summary
str(survey)
```

```
## 'data.frame':    20 obs. of  6 variables:
##  $ Respondent       : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Sex              : num  2 2 1 2 2 2 2 2 2 2 ...
##  $ FathersOccupation: num  1 3 3 3 1 2 3 1 1 1 ...
##  $ PersonsAtHome    : num  5 7 3 8 5 9 6 7 8 4 ...
##  $ SiblingsAtSchool : num  6 4 4 1 2 1 5 3 1 2 ...
##  $ TypesOfHouses    : num  1 2 3 1 1 3 3 1 2 3 ...
```

```
summary(survey)
```

```
##    Respondent          Sex       FathersOccupation PersonsAtHome
##  Min.   : 1.00   Min.   :1.00   Min.   :1.00      Min.   : 3.0
##  1st Qu.: 5.75   1st Qu.:2.00   1st Qu.:1.00      1st Qu.: 5.0
##  Median :10.50   Median :2.00   Median :2.00      Median : 7.0
##  Mean   :10.50   Mean   :1.85   Mean   :1.95      Mean   : 6.4
##  3rd Qu.:15.25   3rd Qu.:2.00   3rd Qu.:3.00      3rd Qu.: 8.0
##  Max.   :20.00   Max.   :2.00   Max.   :3.00      Max.   :11.0
##  SiblingsAtSchool TypesOfHouses
##  Min.   :1.00     Min.   :1.0
##  1st Qu.:2.00     1st Qu.:2.0
##  Median :2.50     Median :2.5
##  Mean   :2.95     Mean   :2.3
##  3rd Qu.:4.25     3rd Qu.:3.0
##  Max.   :6.00     Max.   :3.0
```

```
# Part b: dimensions
dim(survey)
```

```
## [1] 20  6
```

```
nrow(survey)
```

```
## [1] 20
```

```
ncol(survey)
```

```
## [1] 6
```

```
# Part c: factors (stored in separate objects so survey stays numeric)
sex_f <- factor(survey$Sex, levels = c(1, 2), labels = c("Male", "Female"))
father_f <- factor(survey$FathersOccupation, levels = c(1, 2, 3),
                   labels = c("Farmer", "Driver", "Others"))
house_f <- factor(survey$TypesOfHouses, levels = c(1, 2, 3),
                  labels = c("Wood", "Semi-Concrete", "Concrete"))

table(sex_f)
```

```
## sex_f
##   Male Female
##      3     17
```

```
table(father_f)
```

```
## father_f
## Farmer Driver Others
##      9      3      8
```

```
table(house_f)
```

```
## house_f
##          Wood Semi-Concrete      Concrete
##             4             6            10
```

```
# Part d: mean of all columns
colMeans(survey)
```

```
##        Respondent               Sex FathersOccupation     PersonsAtHome
##             10.50              1.85              1.95              6.40
##  SiblingsAtSchool     TypesOfHouses
##              2.95              2.30
```

```
# Part e: first five rows, all columns
survey[1:5, ]
```

```
##   Respondent Sex FathersOccupation PersonsAtHome SiblingsAtSchool TypesOfHouses
## 1          1   2                 1             5                6            1
## 2          2   2                 3             7                4            2
## 3          3   1                 3             3                4            3
## 4          4   2                 3             8                1            1
## 5          5   2                 1             5                2            1
```

```
# Part f: select TypesOfHouses and store it as types_houses
types_houses <- survey$TypesOfHouses
types_houses
```

```
##  [1] 1 2 3 1 1 3 3 1 2 3 2 3 2 2 3 3 3 3 3 2
```

```
# Part g: male respondents whose father was a farmer
survey[survey$Sex == 1 & survey$FathersOccupation == 1, ]
```

```
## [1] Respondent        Sex               FathersOccupation PersonsAtHome
## [5] SiblingsAtSchool  TypesOfHouses
## <0 rows> (or 0-length row.names)
```

```
# Part h: female respondents with 5 or more siblings in school
survey[survey$Sex == 2 & survey$SiblingsAtSchool >= 5, ]
```

```
##    Respondent Sex FathersOccupation PersonsAtHome SiblingsAtSchool
## 1           1   2                 1             5                6
## 7           7   2                 3             6                5
## 13         13   2                 1             4                5
## 14         14   2                 3             7                5
## 18         18   2                 1            11                5
##    TypesOfHouses
## 1              1
## 7              3
## 13             2
## 14             2
## 18             3
```

#Question #2

```
# Answer: Every row is one respondent, so every column has to have an answer
# for the same 20 people. If one vector is shorter, the answers would shift
# and get matched to the wrong respondent. data.frame() would also give an
# error about differing number of rows.
```

#Question #3

```
# Answer: Sex, FathersOccupation, and TypesOfHouses are categories (nominal
# data), because the numbers are just codes. PersonsAtHome and SiblingsAtSchool
# are real counts. Respondent is only an ID number. Averaging Sex would be
# misleading because 1 and 2 are not amounts, so a mean of 1.85 does not mean
# anything.
```

#Question #4

```
survey[survey$Sex == 1, ]
```

```
##    Respondent Sex FathersOccupation PersonsAtHome SiblingsAtSchool
## 3           3   1                 3             3                4
## 11         11   1                 3             7                3
## 19         19   1                 2             7                3
##    TypesOfHouses
## 3              3
## 11             2
## 19             3
```

```
# Answer: The output only shows rows with a 1 in Sex. Someone who has not seen
# the legend would not know that 1 means Male, so they could think it means
# Female or something else. This shows that coded data is hard to read until
# it is labeled, and it is easy to misread it.
```

#Question #5

```
nrow(survey[survey$Sex == 1, ])
```

```
## [1] 3
```

```r
sum(survey$Sex == 1)
```

```
## [1] 3
```

```r
# Answer: There are 3 males out of 20 respondents. This sample is small and
# has very few males, so I would not be confident generalizing any pattern to
# a larger population. A pattern from only 3 people could easily be random.
```

#Question #6

```r
summary(survey)
```

```
##    Respondent         Sex       FathersOccupation PersonsAtHome
##  Min.   : 1.00   Min.   :1.00   Min.   :1.00      Min.   : 3.0
##  1st Qu.: 5.75   1st Qu.:2.00   1st Qu.:1.00      1st Qu.: 5.0
##  Median :10.50   Median :2.00   Median :2.00      Median : 7.0
##  Mean   :10.50   Mean   :1.85   Mean   :1.95      Mean   : 6.4
##  3rd Qu.:15.25   3rd Qu.:2.00   3rd Qu.:3.00      3rd Qu.: 8.0
##  Max.   :20.00   Max.   :2.00   Max.   :3.00      Max.   :11.0
##  SiblingsAtSchool TypesOfHouses
##  Min.   :1.00     Min.   :1.0
##  1st Qu.:2.00     1st Qu.:2.0
##  Median :2.50     Median :2.5
##  Mean   :2.95     Mean   :2.3
##  3rd Qu.:4.25     3rd Qu.:3.0
##  Max.   :6.00     Max.   :3.0
```

```r
# Answer: R does not know the columns are categories because they are stored
# as plain numbers, so it treats them like any other numeric column. The
# meaning of the numbers only comes from the legend, which R does not have.
# So how data is stored (numeric) is different from what it means
# (categories). To fix this, I need to turn them into factors.
```

#Question #7

```r
mean(survey$PersonsAtHome)
```

```
## [1] 6.4
```

```r
mean(survey$Sex)
```

```
## [1] 1.85
```

```r
# Answer: The mean of PersonsAtHome (6.4) is meaningful because it is a real
# count, so it means the average household has about 6 persons. The mean of
# Sex (1.85) is not meaningful because Sex is a category. It only shows that
# most respondents are coded 2, which is better shown with a count or table.
```